

A. Project Overview

- **Goal:** Identify the busiest intersections in the City of Boston road network by computing and comparing three centrality measures—degree, betweenness, and closeness—on a graph of managed streets.
- **Dataset:** City of Boston Managed Streets CSV (contains shape_wkt MultiLineString geometry and STREETNAME columns). Size: ~100,000 segments (link: <https://data.boston.gov/dataset/city-of-boston-managed-streets/resource/4fc84110-6cf8-476d-a566-a9db5f8b3225>)

B. Data Processing

- **Loading into Rust:** Used the csv crate to read rows, located shape_wkt and STREETNAME columns by header index.
- **Cleaning & Transformation:**
 - Trimmed and skipped any rows where the street name was empty.
 - Parsed WKT strings with wkt → converted to geo_types::Geometry.
 - Extracted only MultiLineString geometries, splitting each linestring into a Segment struct carrying (lon, lat) endpoints and name.

C. Code Structure

Modules

- **io.rs:** Handles CSV I/O and WKT parsing into Segment structs.
- **graph.rs:** Builds an undirected Graph of intersections and maps each NodeIndex to its incident street names.
- **centrality.rs:** Computes three centrality measures and prints the top-K true intersections.
- **main.rs:** Orchestrates CLI parsing, data loading, graph assembly, centrality computation, and output.

Key Functions & Types

io::load_segments

- **Purpose:** Read CSV and convert valid WKT rows into Vec<Segment>
- **Inputs:** &Path to CSV

- **Outputs:** Result<Vec<Segment>, Error>
- **Logic:** Locate columns, skip empty names, parse WKT, iterate MultiLineString, collect endpoints.

graph::build_graph

- **Purpose:** Create Graph and node_streets map
- **Inputs:** &[amp;Segment]
- **Outputs:** (Graph, HashMap<NodeIndex, HashSet<String>>)
- **Logic:** Round coords to 6 decimals, de-duplicate via coord_to_node, add nodes & edges, record street names per node.

centrality::degree_centrality

- **Purpose:** Compute normalized degree of each node
- **Inputs:** &Graph
- **Outputs:** HashMap<NodeIndex, f64>
- **Logic:** For each node, count neighbors(v) and divide by n-1.

centrality::betweenness_centrality

- **Purpose:** Compute Brandes' betweenness for unweighted graph
- **Inputs:** &Graph
- **Outputs:** HashMap<NodeIndex, f64>
- **Logic:** For each source s, run BFS to compute sigma, preds, then accumulate dependency delta, divide by 2 for undirected.

centrality::closeness_centrality

- **Purpose:** Compute closeness as (reachable-1)/sum(distances)
- **Inputs:** &Graph
- **Outputs:** HashMap<NodeIndex, f64>
- **Logic:** Dijkstra from each node, sum distances & reachable count, compute formula.

centrality::print_top_with_names

- **Purpose:** Display top-K intersections
- **Inputs:** title, scores map, k, &Graph, &node_streets
- **Outputs:** Printed lines to stdout
- **Logic:** Filter out leaves and single-street nodes, sort by score, print node index, coords, score, joined street names.

Main Workflow

1. **CLI parsing** (main.rs) → get CSV path
2. **Segment loading** (io::load_segments) → Vec<Segment>
3. **Graph building** (graph::build_graph) → Graph + node_streets
4. **Centrality computation** (three functions) → score maps
5. **Output** (print_top_with_names) → top intersections for each measure

D. Tests

```
running 3 tests
test graph::tests::test_build_graph_simple ... ok
test centrality::tests::test_degree_and_closeness_line ... ok
test io::tests::test_parse_named_wkt ... ok

test result: ok. 3 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s
```

- **test_parse_named_wkt:** Verifies MultiLineString parsing.
- **test_build_graph_simple:** Checks node/edge counts and street-name mapping on a tiny graph.
- **test_degree_and_closeness_line:** Validates centrality formulas on a 3-node line.

E. Results

```
--- Degree Centrality ---
Node 232 @ (-71.081748,42.340606): 0.00084 Streets: MASSACHUSETTS AVENUE & COLUMBUS AVENUE
Node 1405 @ (-71.057017,42.329511): 0.00074 Streets: DORCHESTER STREET & DORCHESTER AVENUE & BOSTON STREET & SOUTHAMPTON STREET & PREBLE STREET
Node 1186 @ (-71.008046,42.387120): 0.00063 Streets: FORD STREET & BOARDMAN STREET & SARATOGA STREET
Node 2529 @ (-71.056594,42.297289): 0.00063 Streets: NEPONSET AVENUE & PARKMAN STREET & ADAMS STREET
Node 5764 @ (-71.058009,42.352350): 0.00063 Streets: ESSEX STREET & JOHN F FITZGERALD SURFACE ROAD & LINCOLN STREET
Node 546 @ (-71.070707,42.340790): 0.00063 Streets: PELHAM STREET & WEST DEDHAM STREET & MONSIGNOR REYNOLDS WAY & WASHINGTON STREET
Node 3835 @ (-71.063198,42.373941): 0.00063 Streets: HARVARD STREET & PLEASANT STREET & PRESCOTT STREET & MAIN STREET & DEVENS STREET
Node 913 @ (-71.071781,42.348637): 0.00063 Streets: COLUMBUS AVENUE & BERKELEY STREET
Node 3555 @ (-71.006795,42.386664): 0.00063 Streets: SARATOGA STREET & BENNINGTON STREET
Node 125 @ (-71.058532,42.366921): 0.00063 Streets: ENDICOTT STREET & NORTH WASHINGTON STREET & CAUSEWAY STREET
Node 1841 @ (-71.081923,42.321727): 0.00053 Streets: WARREN STREET & COPELAND STREET & WAVERLY STREET
Node 3248 @ (-71.093701,42.273313): 0.00053 Streets: BLUE HILL AVENUE & WOODHAVEN STREET
Node 4177 @ (-71.073641,42.299371): 0.00053 Streets: HARVARD STREET & BOWDOIN AVENUE & BOWDOIN STREET & WASHINGTON STREET
Node 2113 @ (-71.137227,42.280696): 0.00053 Streets: CORNELL STREET & WALMORTH STREET & LENISTON STREET
Node 1830 @ (-71.100092,42.335829): 0.00053 Streets: LONGWOOD AVENUE & HUNTINGTON AVENUE
Node 6677 @ (-71.110312,42.310232): 0.00053 Streets: NEMBERN STREET & REVERE STREET & ROANOKE AVENUE & ELM STREET
Node 3164 @ (-71.068362,42.312682): 0.00053 Streets: GLENDALE STREET & COLUMBIA ROAD & BELLEVUE STREET
Node 654 @ (-71.085100,42.328899): 0.00053 Streets: WASHINGTON STREET & MALCOLM X BOULEVARD & DUDLEY STREET
Node 1279 @ (-71.083161,42.308318): 0.00053 Streets: WASHINGTON STREET & BLUE HILL AVENUE
Node 2720 @ (-71.167607,42.350572): 0.00053 Streets: OAK SQUARE & WASHINGTON STREET
--- Betweenness Centrality ---
Node 5644 @ (-71.118871,42.279930): 4445966.34858 Streets: CUMMINS HIGHWAY & HYDE PARK AVENUE
Node 1521 @ (-71.118969,42.279972): 4077276.82145 Streets: HYDE PARK AVENUE & CUMMINS HIGHWAY
Node 4316 @ (-71.122433,42.281826): 4068248.35374 Streets: BROWN AVENUE & CUMMINS HIGHWAY
Node 5357 @ (-71.106998,42.310463): 3823372.37745 Streets: GREEN STREET & AMORY STREET
Node 5458 @ (-71.116216,42.296068): 3690143.21320 Streets: WALK HILL STREET & HYDE PARK AVENUE
Node 2745 @ (-71.117717,42.292436): 3609663.49690 Streets: ELDRIDGE ROAD & HYDE PARK AVENUE
Node 438 @ (-71.105624,42.312624): 3556260.85326 Streets: AMORY STREET & CORNMALL STREET
Node 2479 @ (-71.106382,42.307588): 3546762.77895 Streets: UNION AVENUE & WASHINGTON STREET
Node 225 @ (-71.106603,42.307364): 3544819.19863 Streets: WASHINGTON STREET & KENTON ROAD
Node 4732 @ (-71.118120,42.289879): 3539285.34498 Streets: HYDE PARK AVENUE & SOUTHBOURNE ROAD
Node 7456 @ (-71.109110,42.304830): 3534471.96740 Streets: BROOKLEY ROAD & WASHINGTON STREET
Node 224 @ (-71.107023,42.306947): 3531722.69863 Streets: WASHINGTON STREET & GARTLAND STREET
Node 6305 @ (-71.108438,42.305511): 3524251.19863 Streets: MCBRIDE STREET & WASHINGTON STREET
Node 2241 @ (-71.118574,42.286755): 3520419.14535 Streets: HYDE PARK AVENUE & FLORIAN STREET
Node 6982 @ (-71.109271,42.304675): 3517947.51933 Streets: WASHINGTON STREET & BURNETT STREET
Node 1128 @ (-71.074864,42.331268): 3516580.16593 Streets: MELNEA CASS BOULEVARD & HAMPDEN STREET
Node 4361 @ (-71.110077,42.303817): 3515496.54625 Streets: BURNETT STREET & WASHINGTON STREET
Node 4837 @ (-71.073513,42.333509): 3513758.46482 Streets: MASSACHUSETTS AVENUE & ALBANY STREET
Node 2924 @ (-71.072290,42.332615): 3512513.16224 Streets: MASSACHUSETTS AVENUE & MELNEA CASS BOULEVARD
Node 1672 @ (-71.113545,42.299940): 3509150.85076 Streets: TOWER STREET & WASHINGTON STREET & HYDE PARK AVENUE
--- Closeness Centrality ---
Node 9350 @ (-71.012704,42.381601): 1.00000 Streets: COLERIDGE STREET & RICE STREET
Node 1803 @ (-71.062394,42.316735): 1.00000 Streets: KEVIN ROAD & SUMNER PARK
Node 7755 @ (-71.115798,42.351017): 1.00000 Streets: SAINT PAUL STREET & BUICK STREET
Node 4060 @ (-71.088082,42.276341): 1.00000 Streets: WOODDALE AVENUE & BANFIELD AVENUE
Node 6203 @ (-71.053977,42.290827): 1.00000 Streets: MANOR STREET & MATIGNON ROAD
Node 2225 @ (-71.105999,42.346255): 1.00000 Streets: KESWICK STREET & BEACON STREET
Node 976 @ (-71.054487,42.318963): 1.00000 Streets: RAVEN STREET & CARSON STREET
Node 728 @ (-71.086299,42.342867): 1.00000 Streets: PUBLIC ALLEY NO 801 & PUBLIC ALLEY NO 818
Node 261 @ (-71.049300,42.283193): 1.00000 Streets: MARSH STREET & GALLIVAN BOULEVARD
Node 7491 @ (-71.125029,42.240218): 1.00000 Streets: BADGER ROAD & CRANMORE ROAD
Node 6117 @ (-71.055833,42.339688): 1.00000 Streets: EAST FIFTH STREET & WEST FIFTH STREET
Node 7652 @ (-71.139952,42.255434): 1.00000 Streets: KOVEY COURT & KOVEY ROAD
Node 1751 @ (-71.052387,42.333897): 1.00000 Streets: LARK STREET & LORING STREET
Node 5888 @ (-71.134258,42.346576): 1.00000 Streets: COLUMBIA STREET & WALBRIDGE STREET
Node 3208 @ (-71.062157,42.334543): 1.00000 Streets: WIDETT CIRCLE & FOODMART ROAD
Node 4649 @ (-71.083173,42.335343): 1.00000 Streets: GREENWICH STREET & WESTMINSTER STREET
Node 3439 @ (-71.122631,42.308393): 1.00000 Streets: CENTRE STREET & HILLCROFT ROAD
Node 5552 @ (-71.070997,42.316142): 1.00000 Streets: OLEANDER STREET & HALF MOON STREET
Node 7055 @ (-71.073586,42.272492): 1.00000 Streets: MONSON STREET & CADDY ROAD
Node 8966 @ (-71.085789,42.341320): 1.00000 Streets: PUBLIC ALLEY NO 821 & PUBLIC ALLEY NO 822
```

- **Interpretation:** Top degree intersections are high-traffic crossroads in downtown. Betweenness highlights major through-routes (Cummins Hwy, Hyde Park Ave).

F. Usage Instructions

1. Build & Test

cargo clean

```
cargo build --release
```

```
cargo test
```

2. Run

```
cargo run --release -- "<path >"
```

- **Argument:** full path to the CSV file
- **Output:** printed top-20 intersections for each centrality

3. Runtime: ~20s on a modern laptop for the entire data set.